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Assignment No 1

Q1]

Ans]

```
#include<stdio.h>
#include<string.h>
#define Ns

struct weather{
    int id
    char location[30], time[30];
    Float temp, humidity;
} q[N];

int front=-1, rear=-1, count=0;

void enqueue(){
 if(count==N){ printf("Queue Full:\n"); return;}
 rear =(rear+1)%N;
 printf("%d %29s %29s %f ", &q[read.id, q[rear].
location, q[rear].time, &q[rear].temp, &q[rear].
 humidity);
if(front==-1) front=0;
 count++;
 pnntf("Observation Added:\n");
}
void display (int i){
 printf("ID:%d location: %s Time= %if
 Humidity :%.if %%\n", q[i].id, q[i].location,
 q[i].time, q[i].temp, q[i].humidity);
}
```

```
void dequeue(){
 if(!count) {printf("Queue Empty !\n"); return;}
 printf("Removed:"); display(front);
 front = (front+1) %N;
 if(--count==0){front=rear=-1;
 }
void peek(){
 if(!count){printf("Queue Empty !\n");}
   else{ display(front);
   }
void show(){
    int i,j;
    if(!count){ printf("Queue Empty !\n"); return;}
    for(i=front, j=0; j<count; j++, i=(i+1)%N){
       display(i);
    }
void search(){
      int id, i, j;
    printf("Enter ID to search:");
    scanf("%d", &id);
    For(i=front j=0; j<count; j++; i=(i+1)%N){
    if(a[i].id==id){ display(i); return;}
    printf("Not found !\n");
    }

int main(){
     int ch;
    do {
    printf("\n 1. Enqueue 2. Dequeue 3. peek
                4. display");
```

```
printf("\n 5.search 6.count 7. Queue status
        8. Exit \n choice: ");
scanf("%d", &ch);

switch (ch) {
    case1: Enqueue(); ; break;
    case2: Dequeue(); break;
    case3: peek(); break;
    case4: show(); break;
    case5: search(); break;
    case6: printf("count = %d \n", count); break
    case7: printf("Queue is %s \n", count==0?
                "Empty" : count==N?
                "Full" : "Not full");
           break;
    case8: printf("Exit \n"); break;
    default: printf("Invalid choice! \n"); }
} while (ch != 8);
  return 0;
```

Time and space complexity

Operations	Time complexity
Enqueue	O(1)
Dequeue	O(1)
Peek	O(1)
Display	O(n)
Search	O(n)
Count	O(1)
Queue status	O(1)